Solution The information is represented in the form of a 3×2 matrix as follows

$$A=\begin{pmatrix}23 & 18\\ 47 & 36\\ 15 & 16\end{pmatrix}$$

The entry in the second row and first column represent that there are 47 men workers in factory II.

Example 3.57 If a matrix has 16 elements, what are the possible orders it can have?

Solution We know that a matrix of order $m\times n$, has mn elements. Thus to find all possible orders of a matrix with 16 elements, we will find all ordered pairs of natural numbers whose product is 16.

Such ordered pairs are $(1,16)$, $(16,1)$, $(4,4)$, $(8,2)$, $(2,8)$

Hence possible orders are $1\times16,\ 16\times1,\ 4\times4,\ 2\times8,\ 8\times2$



Activity 5

No.	Elements	Possible orders	Number of possible orders
1.	4		3
2.		$1\times9,\ 9\times1,\ 3\times3$	
3.	20		
4.	8		4
5.	1		
6.	100		
7.		$1\times10,\ 10\times1,\ 2\times5,\ 5\times2$	

Do you find any relationship between number of elements (second column) and number of possible orders (fourth column)? If so, what is it?

Example 3.58 Construct a 3×3 matrix whose elements are $a_{ij}=i^2j^2$

Solution The general 3×3 matrix is given by $A=\begin{pmatrix}a_{11} & a_{12} & a_{13}\\ a_{21} & a_{22} & a_{23}\\ a_{31} & a_{32} & a_{33}\end{pmatrix}$ $\qquad a_{ij}=i^2j^2$

$a_{11}=1^2\times1^2=1\times1=1;\quad a_{12}=1^2\times2^2=1\times4=4;\quad a_{13}=1^2\times3^2=1\times9=9;$

$a_{21}=2^2\times1^2=4\times1=4;\quad a_{22}=2^2\times2^2=4\times4=16;\quad a_{23}=2^2\times3^2=4\times9=36$

$a_{31}=3^2\times1^2=9\times1=9;\quad a_{32}=3^2\times2^2=9\times4=36;\quad a_{33}=3^2\times3^2=9\times9=81$

Hence the required matrix is $A=\begin{pmatrix}1 & 4 & 9\\ 4 & 16 & 36\\ 9 & 36 & 81\end{pmatrix}$

Example 3.59 Find the value of $a,\ b,\ c,\ d$ from the equation $\begin{pmatrix}a-b & 2a+c\\ 2a-b & 3c+d\end{pmatrix}=\begin{pmatrix}1 & 5\\ 0 & 2\end{pmatrix}$

Solution The given matrices are equal. Thus all corresponding elements are equal.

3 Algebra.indd 143 16-12-2020 01:07:03 PM